

SIMATS ENGINEERING
ASSESSMENT TOOL II – DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : 192524225 Name: VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

Criteria	Problem Understanding & Data Preparation 25%	Data Analysis & Interpretation 30%	Application of Analytical Techniques 20%	Analysis of Results 15%	Organization & Documentation 10%	Presentation & Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							

Q5

Data Analysis Project

Question 1 – Distribution Analysis Using ECDF and Q-Q Plot

Dataset: Employee_Salaries.csv

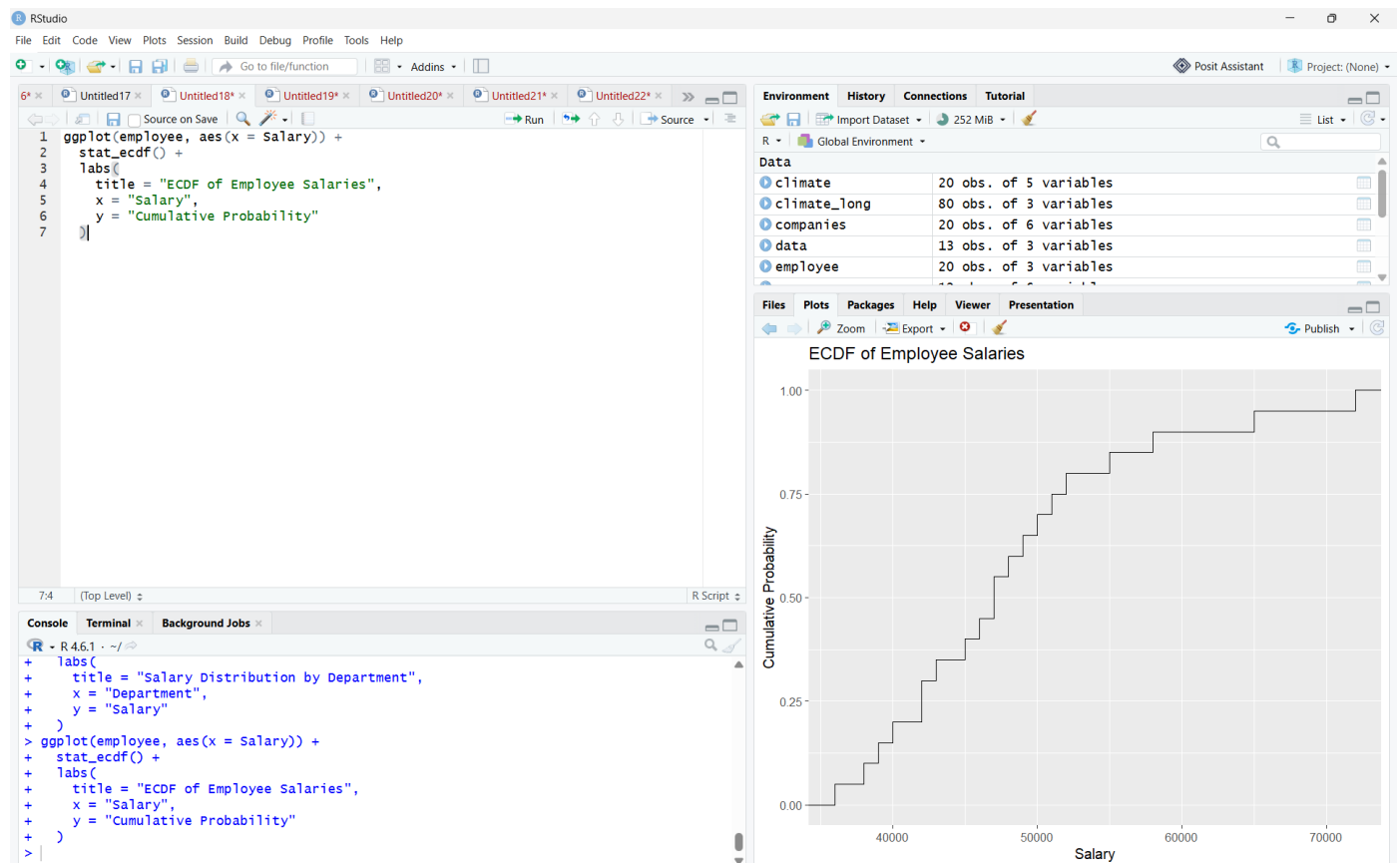
Assessment Task

Using the Employee_Salaries dataset, analyze the distribution of employee salaries.

1. Create an empirical cumulative distribution function (ECDF) for Salary.

Answer:

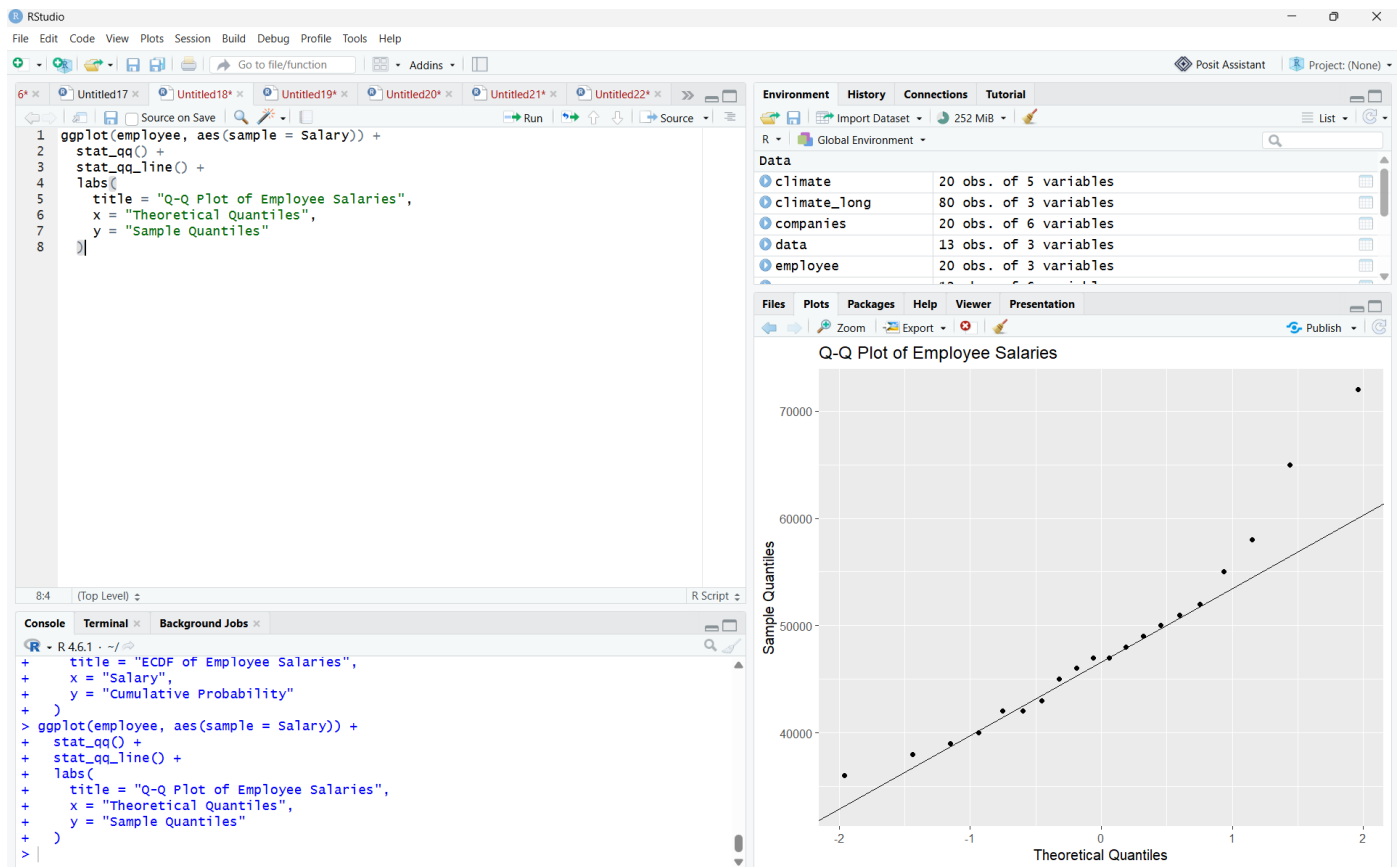
The ECDF displays the cumulative proportion of employees whose salaries are less than or equal to each salary value. As salary increases, the cumulative probability increases from 0 to 1.



2. Create a Q-Q plot to assess whether Salary is approximately normally distributed.

Answer:

The Q-Q plot compares the observed salary quantiles with theoretical normal-distribution quantiles. If the points approximately follow the reference line, the data are approximately normal.



3. Compare the salary distributions of the four departments using suitable distribution plots.

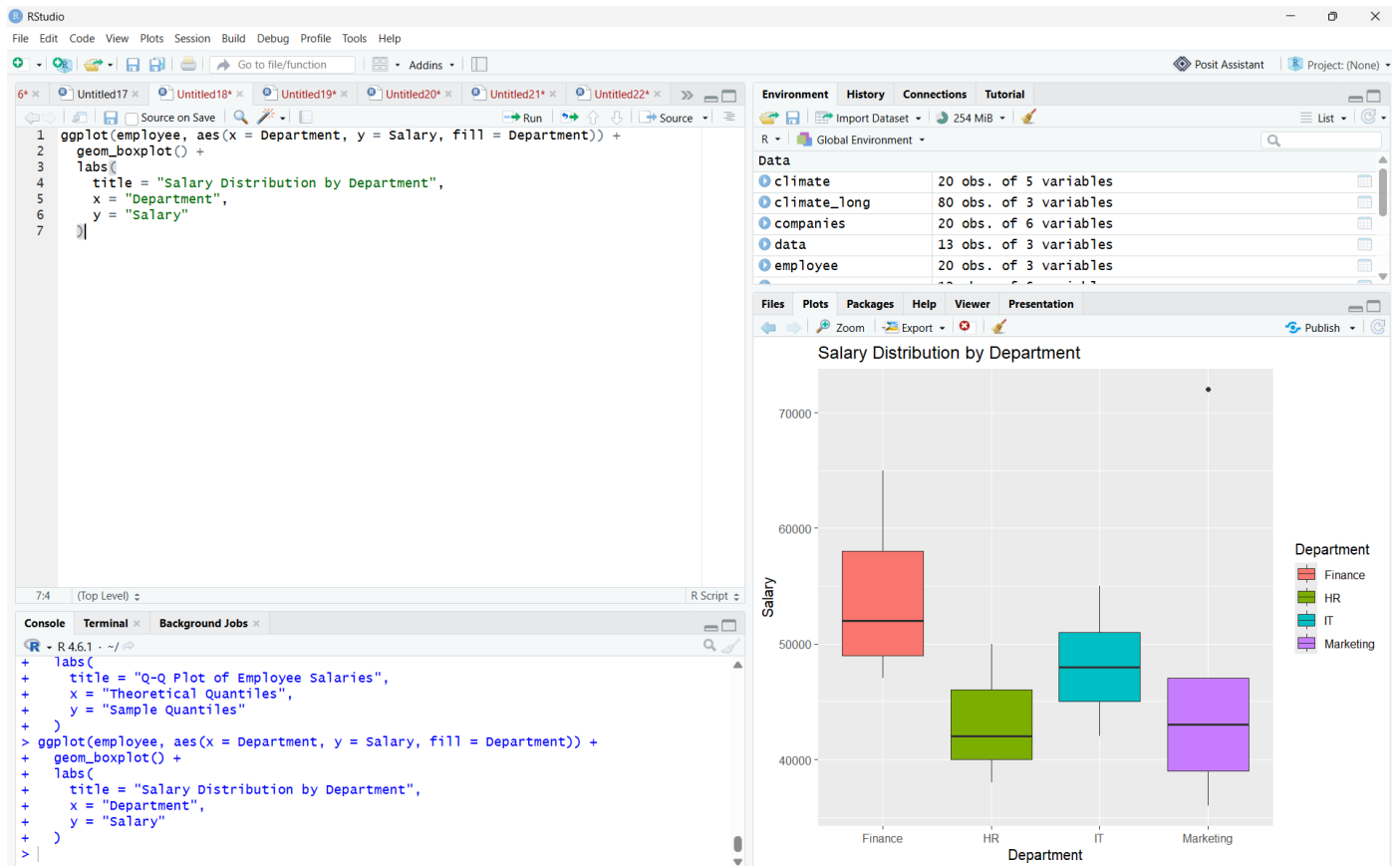
Answer:

The salary distributions can be compared using box plots or violin plots.

Comparison:

- IT: ₹42,000–₹55,000
- HR: ₹38,000–₹50,000
- Finance: ₹47,000–₹65,000
- Marketing: ₹36,000–₹72,000

Finance has relatively high salaries, while Marketing has the widest range.



4. Identify any possible skewness or unusual observations and explain what the plots reveal.

Answer:

The Marketing department contains an unusually high salary of ₹72,000. This observation can produce right-skewness in the Marketing salary distribution.

The ECDF shows cumulative salary probabilities, the Q-Q plot evaluates normality, and the box plot compares salary distributions between departments. Together, the plots show differences in salary level, spread and unusual observations.

Dataset:

Employee_ID	Department	Salary
E001	IT	42000
E002	IT	45000
E003	IT	48000
E004	IT	51000
E005	IT	55000

E006	HR	38000
E007	HR	40000
E008	HR	42000
E009	HR	46000
E010	HR	50000
E011	Finance	47000
E012	Finance	49000
E013	Finance	52000
E014	Finance	58000
E015	Finance	65000
E016	Marketing	36000
E017	Marketing	39000
E018	Marketing	43000
E019	Marketing	47000
E020	Marketing	72000

Question 2 – Visualizing Many Distributions Using Bean Plots

Dataset: Student_Performance.csv

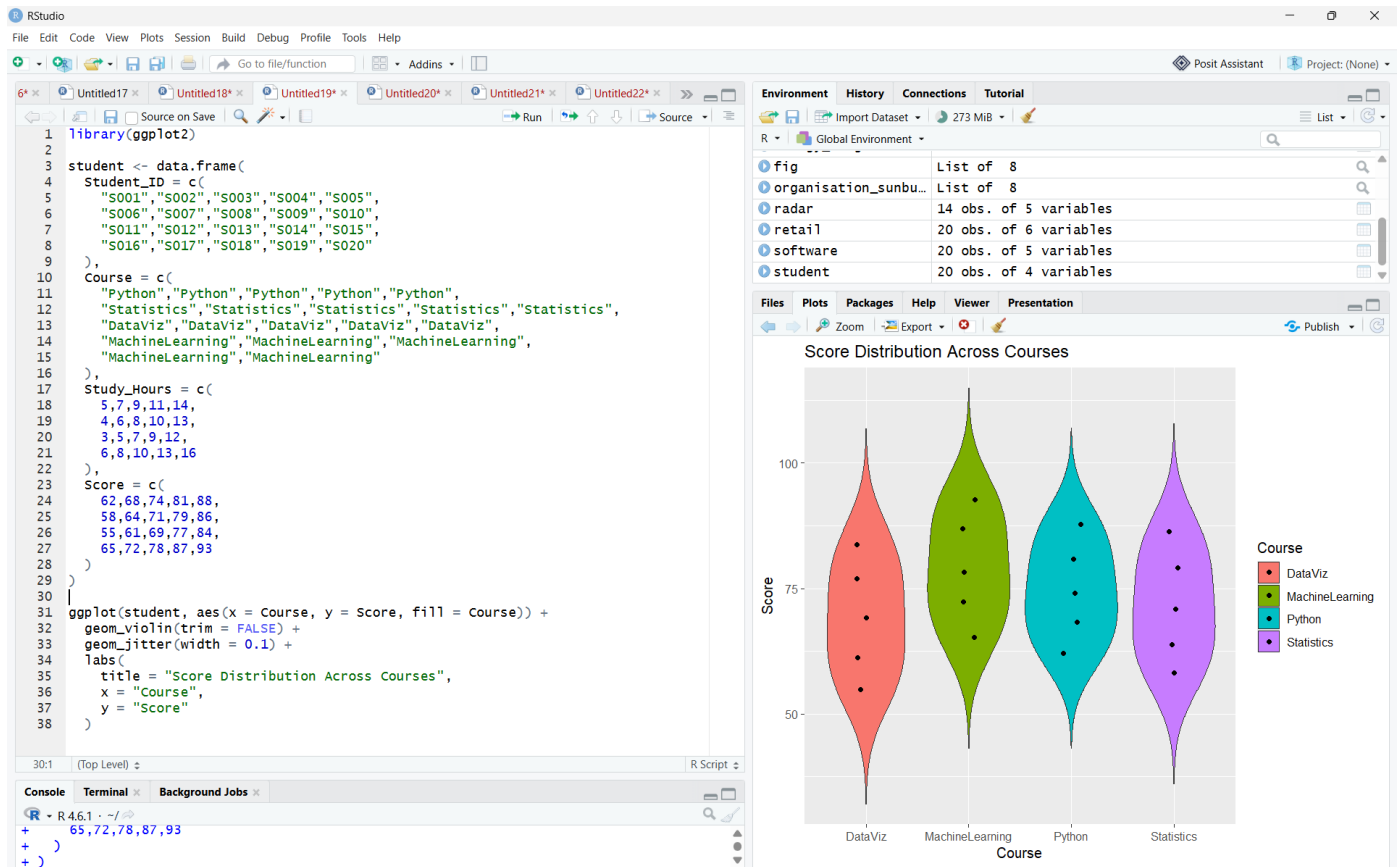
Assessment Task

Using the Student_Performance dataset, investigate how scores are distributed across courses.

1. Visualize the score distribution for all four courses at the same time.

Answer:

This displays the four course score distributions together.



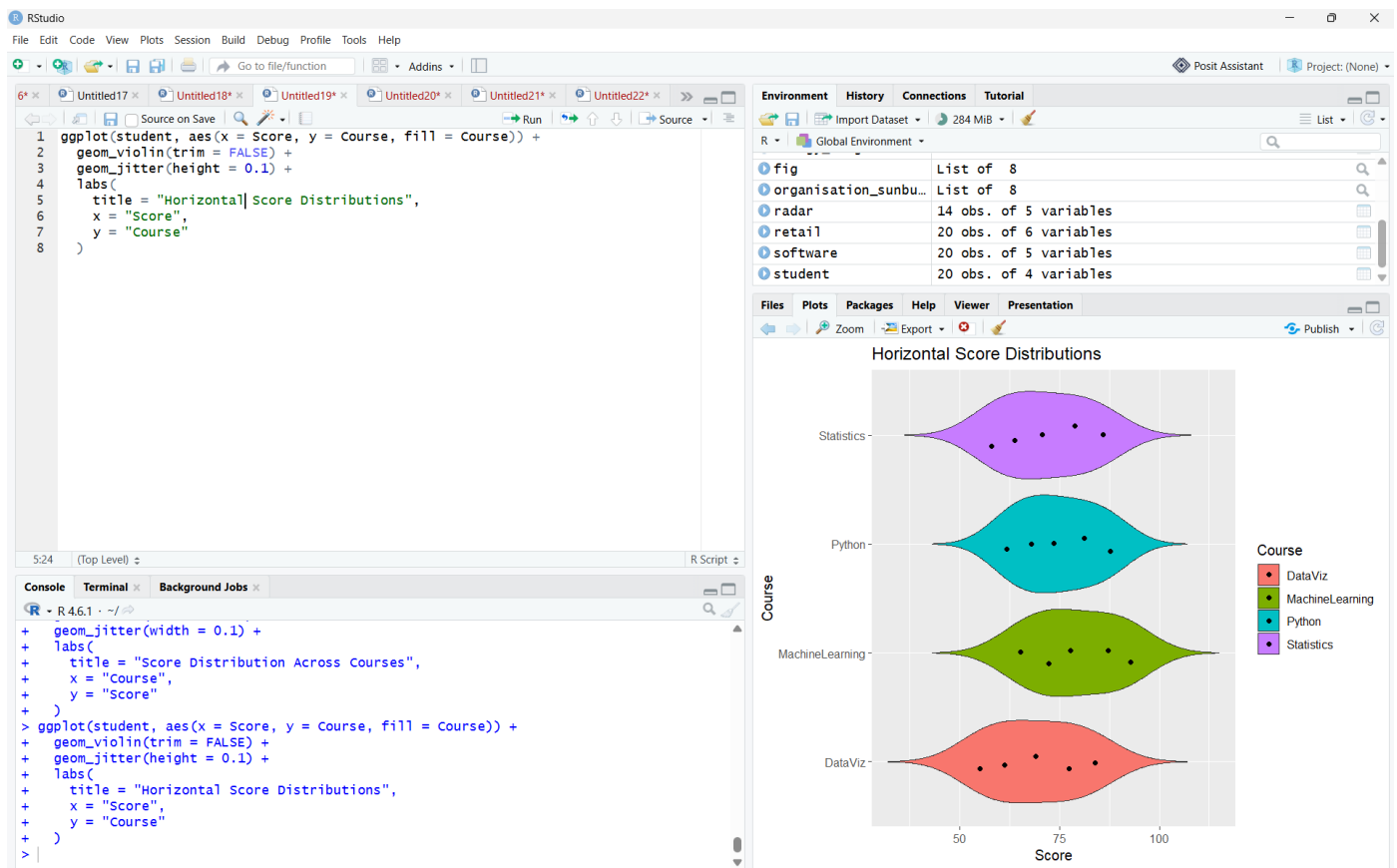
2. Create bean plots (or an equivalent violin/density-based visualization if a bean-plot package is unavailable).

Answer:

The violin plot shows the distribution of scores for all four courses. It displays the center, spread, and shape of the scores. The plot can be used as an equivalent to a bean plot when a bean-plot package is unavailable.

3. Create a second visualization with the distributions arranged along the vertical axis.

Answer:



4. Compare the center, spread, and shape of the distributions and determine which course has the highest and most consistent performance.

Answer:

Mean scores:

Course	Mean
Python	74.6
Statistics	71.6
DataViz	69.2
Machine Learning	79.0

Machine Learning has the highest center with a mean of 79.0. DataViz has the lowest mean of 69.2. The distributions show increasing scores across the given observations for each course.

MachineLearning has the highest average performance with a mean score of **79.0**. It is therefore the best-performing course in the dataset.

Dataset:

Student_ID	Course	Study_Hours	Score
S001	Python	5	62
S002	Python	7	68

S003	Python	9	74
S004	Python	11	81
S005	Python	14	88
S006	Statistics	4	58
S007	Statistics	6	64
S008	Statistics	8	71
S009	Statistics	10	79
S010	Statistics	13	86
S011	DataViz	3	55
S012	DataViz	5	61
S013	DataViz	7	69
S014	DataViz	9	77
S015	DataViz	12	84
S016	MachineLearning	6	65
S017	MachineLearning	8	72
S018	MachineLearning	10	78
S019	MachineLearning	13	87
S020	MachineLearning	16	93

Question 3 – Comparing Proportions with Pie, Side-by-Side and Stacked Bars Dataset: Customer_Purchases.csv

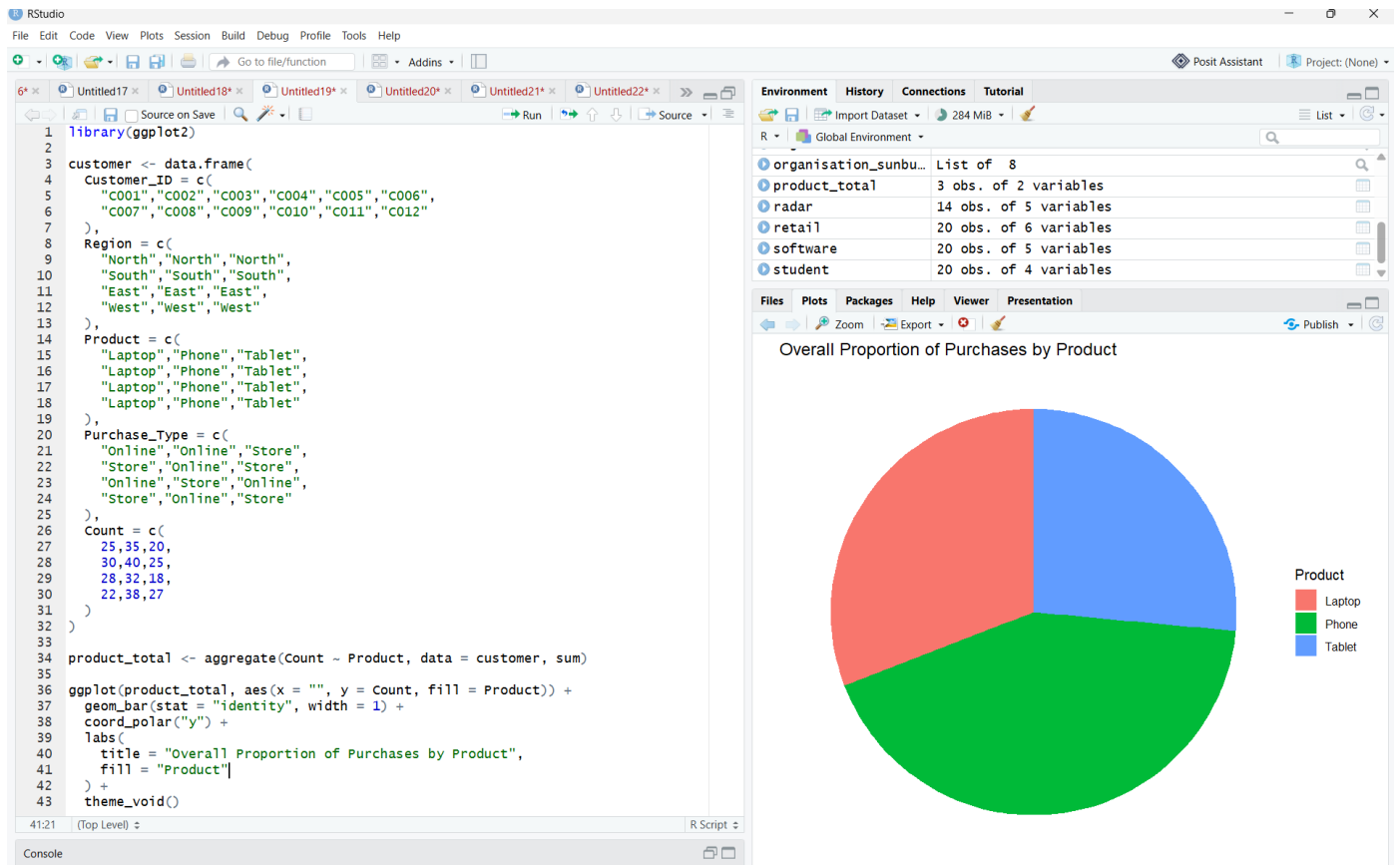
Assessment Task

Using the Customer_Purchases dataset, analyze purchase proportions.

1. Create a pie chart showing the overall proportion of purchases by Product.

Answer:

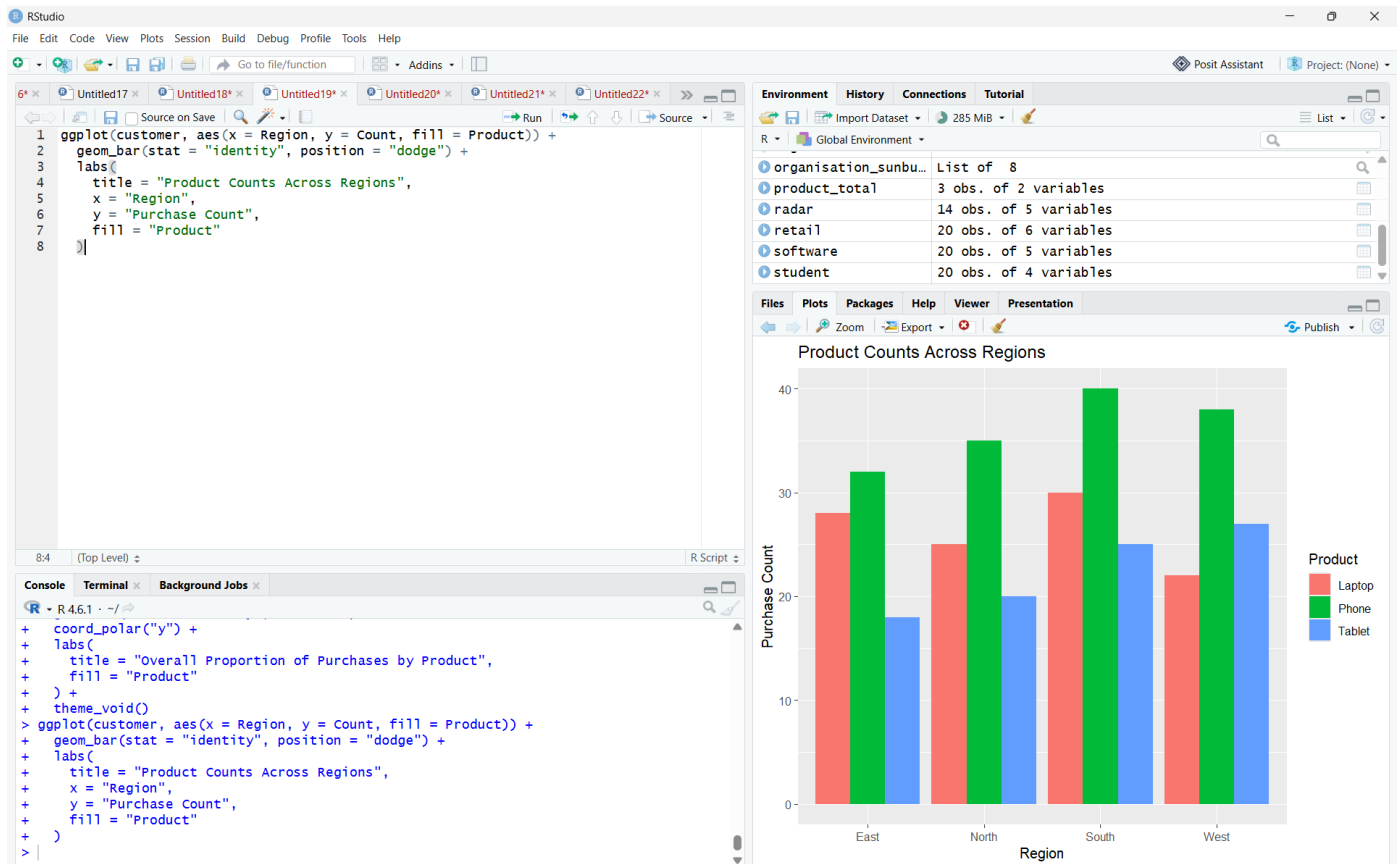
Phone has the highest overall purchase proportion, followed by Laptop and Tablet.



2. Create side-by-side bar charts comparing Product counts across Regions.

Answer:

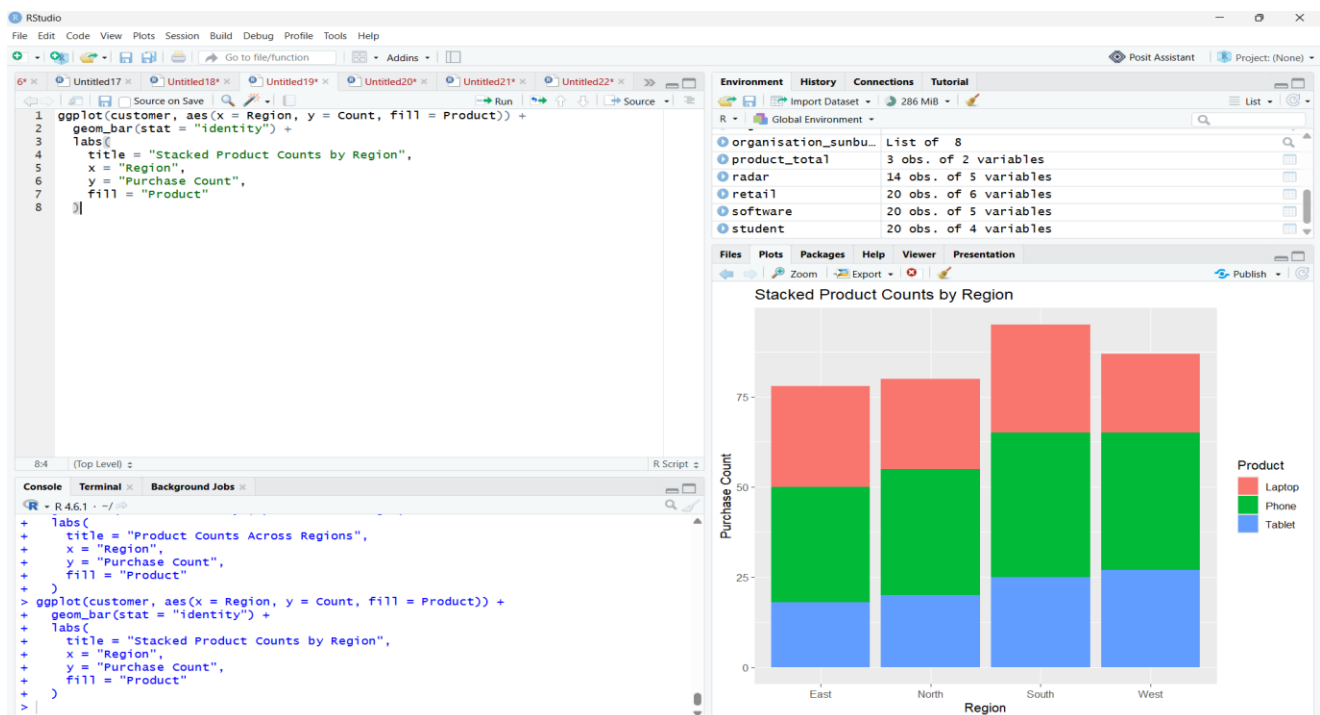
- South has the highest Laptop count: 30.
- South has the highest Phone count: 40.
- West has the highest Tablet count: 27.
- Phone has the highest count in every region.



3. Create a stacked bar chart showing product composition within each Region.

Answer:

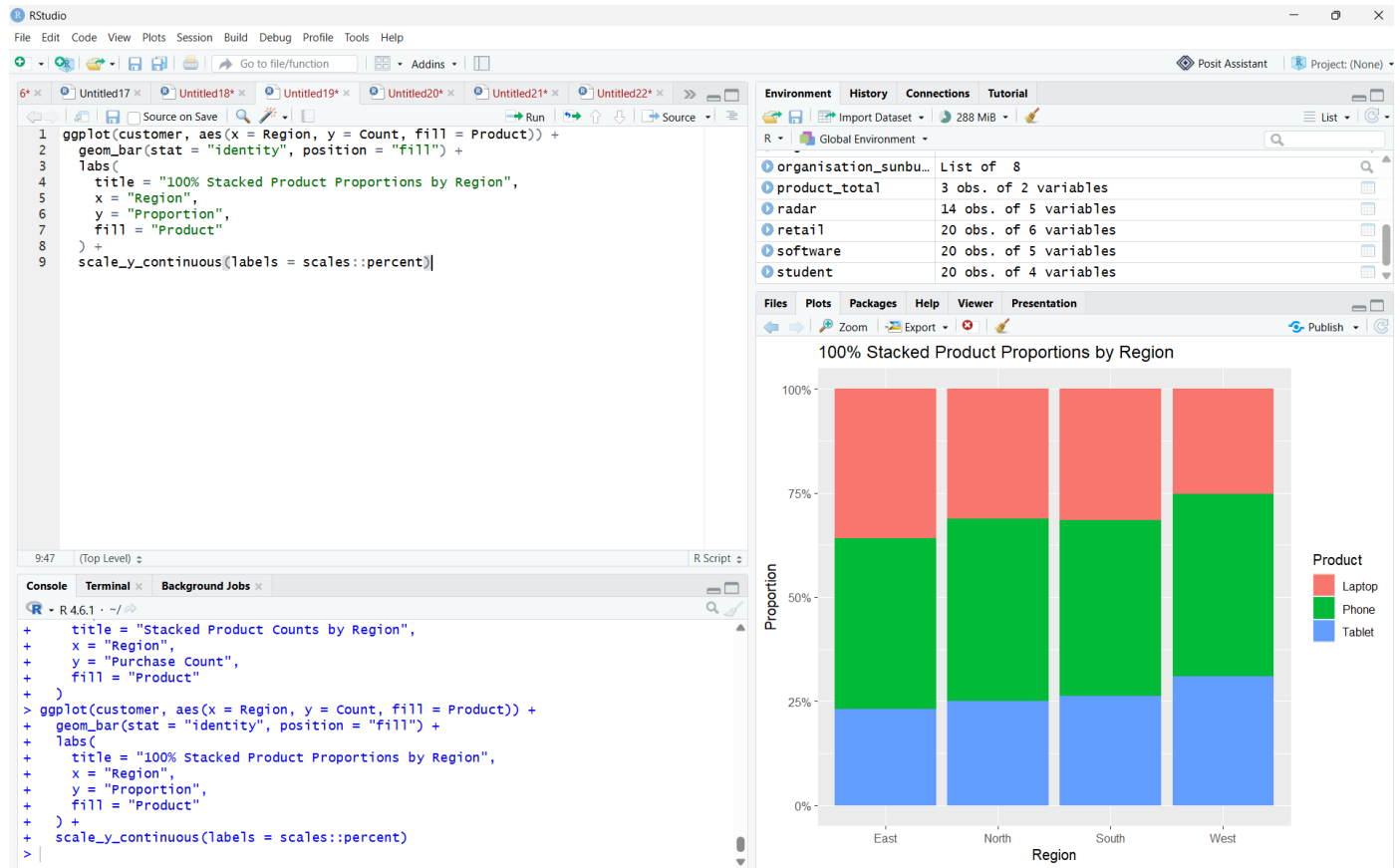
South has the highest total purchase count with 95, while East has the lowest with 78.



4. Convert the stacked chart into a 100% stacked bar chart and compare proportions rather than raw counts.

Answer:

The 100% stacked bar chart converts the raw counts into proportions, so every region has a total height of 100%.



5. Explain why one of these visualizations may be more appropriate than another for comparing regional proportions.

Answer:

The 100% stacked bar chart is the most appropriate visualization for comparing regional proportions.

Explanation

- Every region is normalized to 100%.
- It allows direct comparison of product proportions.
- Differences in regional sample sizes do not dominate the comparison.
- It clearly shows the percentage contribution of each product.
- A pie chart is better for showing overall proportions but not for comparing several regions.
- A side-by-side bar chart is better for comparing raw counts.
- A regular stacked bar chart shows both total size and composition.

Dataset:

Customer_ID	Region	Product	Purchase_Type	Count
C001	North	Laptop	Online	25
C002	North	Phone	Online	35
C003	North	Tablet	Store	20
C004	South	Laptop	Store	30
C005	South	Phone	Online	40
C006	South	Tablet	Store	25
C007	East	Laptop	Online	28
C008	East	Phone	Store	32
C009	East	Tablet	Online	18
C010	West	Laptop	Store	22
C011	West	Phone	Online	38
C012	West	Tablet	Store	27

Question 4 – Nested Proportions Using Treemap and Sunburst Charts

Dataset: Company_Sales.csv

Assessment Task

Using the Company_Sales dataset, visualize nested proportions in the sales hierarchy.

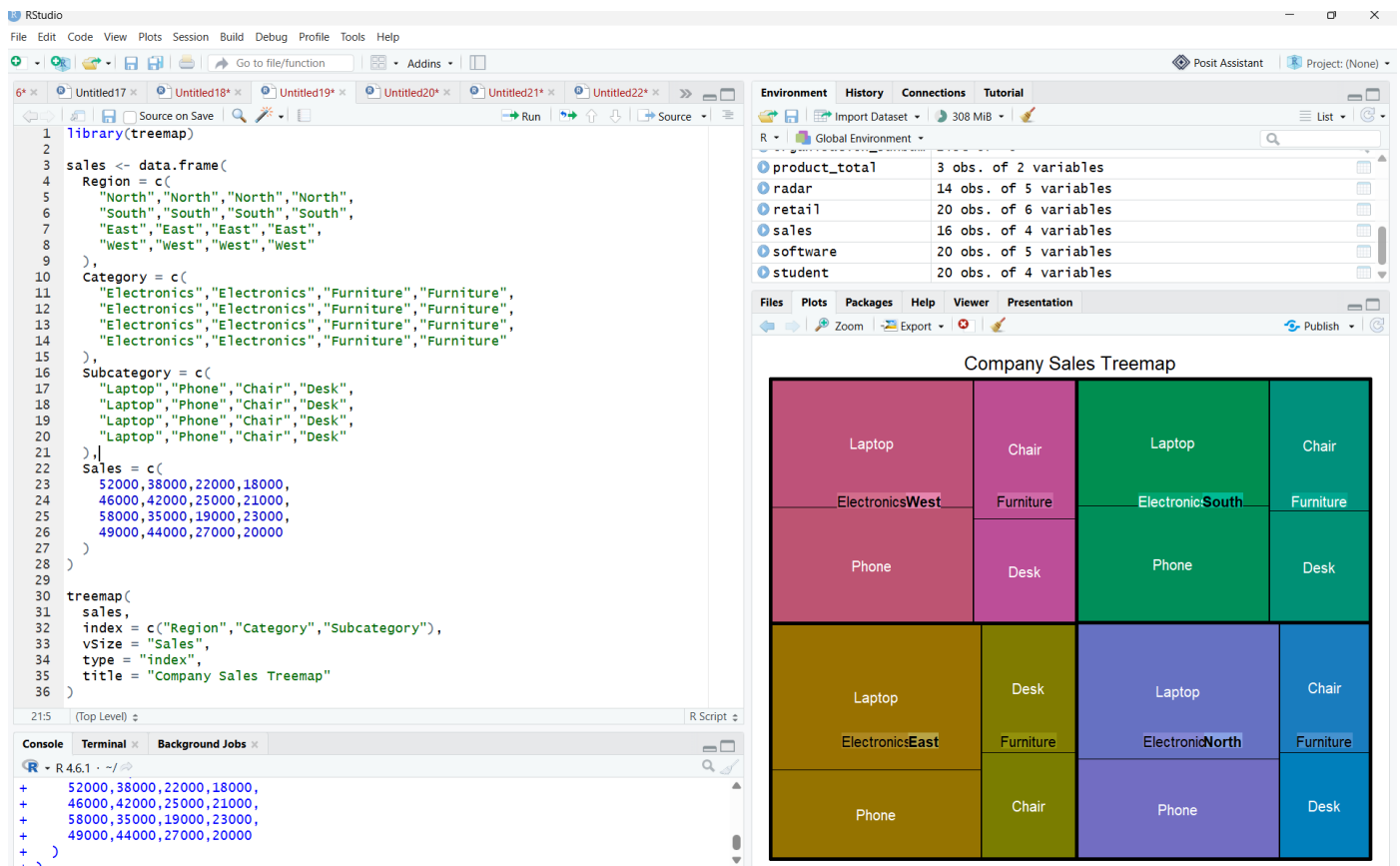
1. Create a treemap showing Region → Category → Subcategory, with rectangle size representing Sales.

Answer:

The treemap represents:

Region → Category → Subcategory

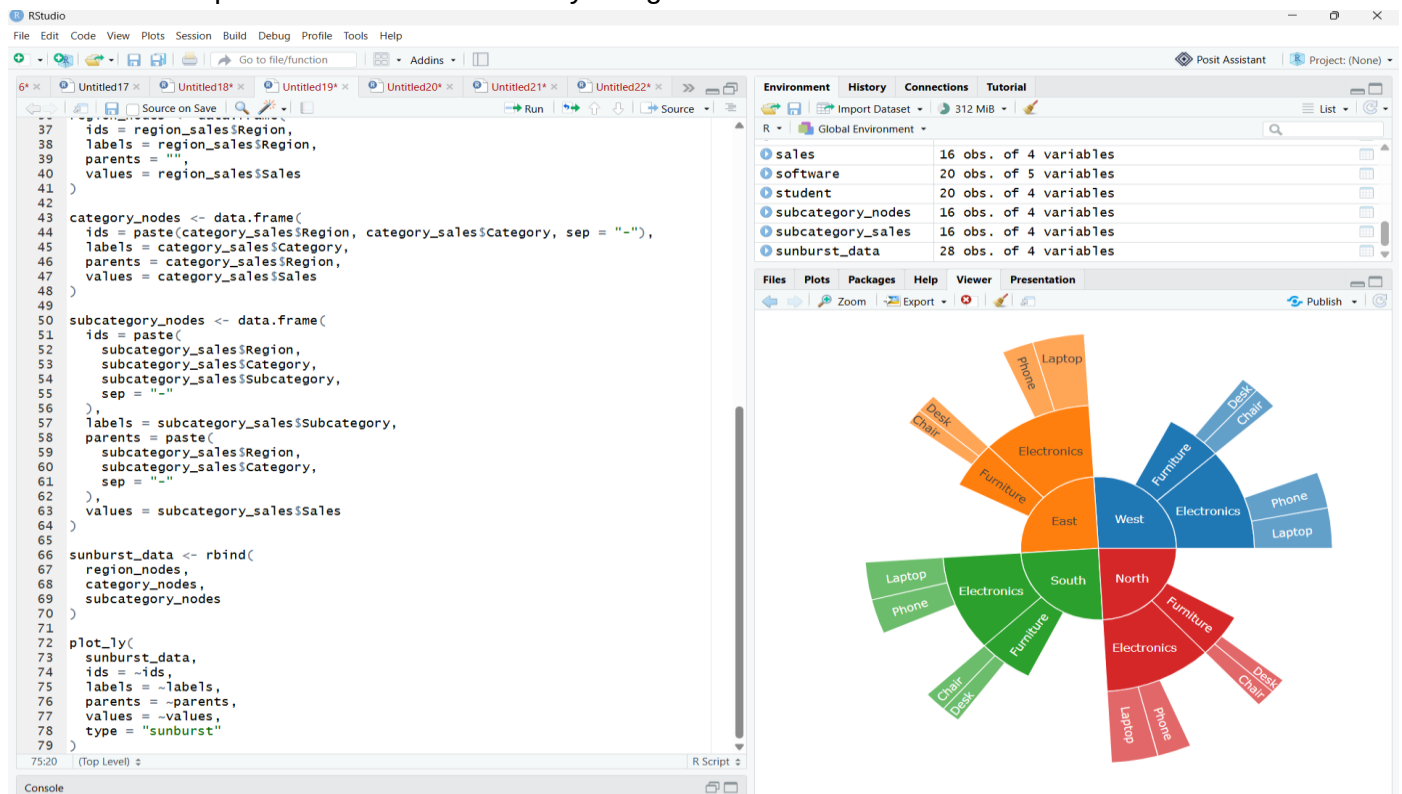
The size of each rectangle represents the corresponding Sales value.



2. Create a sunburst chart representing the same hierarchy.

Answer:

The sunburst represents the same hierarchy using concentric levels.



3. Identify the largest region, category, and subcategory by sales.

Answer:

Region Sales:

Region	Total Sales
North	₹130,000
South	₹134,000
East	₹135,000
West	₹140,000

Category Sales:

Category	Total Sales
Electronics	₹364,000
Furniture	₹175,000

Subcategory Sales

Subcategory	Total Sales
Laptop	₹205,000
Phone	₹159,000
Chair	₹93,000
Desk	₹82,000

- **Largest Region:** West — ₹140,000
- **Largest Category:** Electronics — ₹364,000
- **Largest Subcategory:** Laptop — ₹205,000

4. Compare the readability of the treemap and sunburst chart for identifying nested proportions.

Answer:

Visualization	Readability
Treemap	Easier for comparing sales sizes
Sunburst	Easier for understanding hierarchy

5. State one potential limitation of each visualization.

Answer:

Treemap Limitation

Treemap rectangles with similar sizes can be difficult to compare accurately.

Sunburst Limitation

Comparing values using angles and arc areas can be difficult, especially when many categories are present.

Dataset:

Region	Category	Subcategory	Sales
North	Electronics	Laptop	52000

North	Electronics	Phone	38000
North	Furniture	Chair	22000
North	Furniture	Desk	18000
South	Electronics	Laptop	46000
South	Electronics	Phone	42000
South	Furniture	Chair	25000
South	Furniture	Desk	21000
East	Electronics	Laptop	58000
East	Electronics	Phone	35000
East	Furniture	Chair	19000
East	Furniture	Desk	23000
West	Electronics	Laptop	49000
West	Electronics	Phone	44000
West	Furniture	Chair	27000
West	Furniture	Desk	20000

Question 5 – Flow-Based Proportions with Sankey Diagram and Parallel Sets

Dataset: Student_Career_Path.csv

Assessment Task

Using the Student_Career_Path dataset, analyze how students flow from specialization to internship status and placement sector.

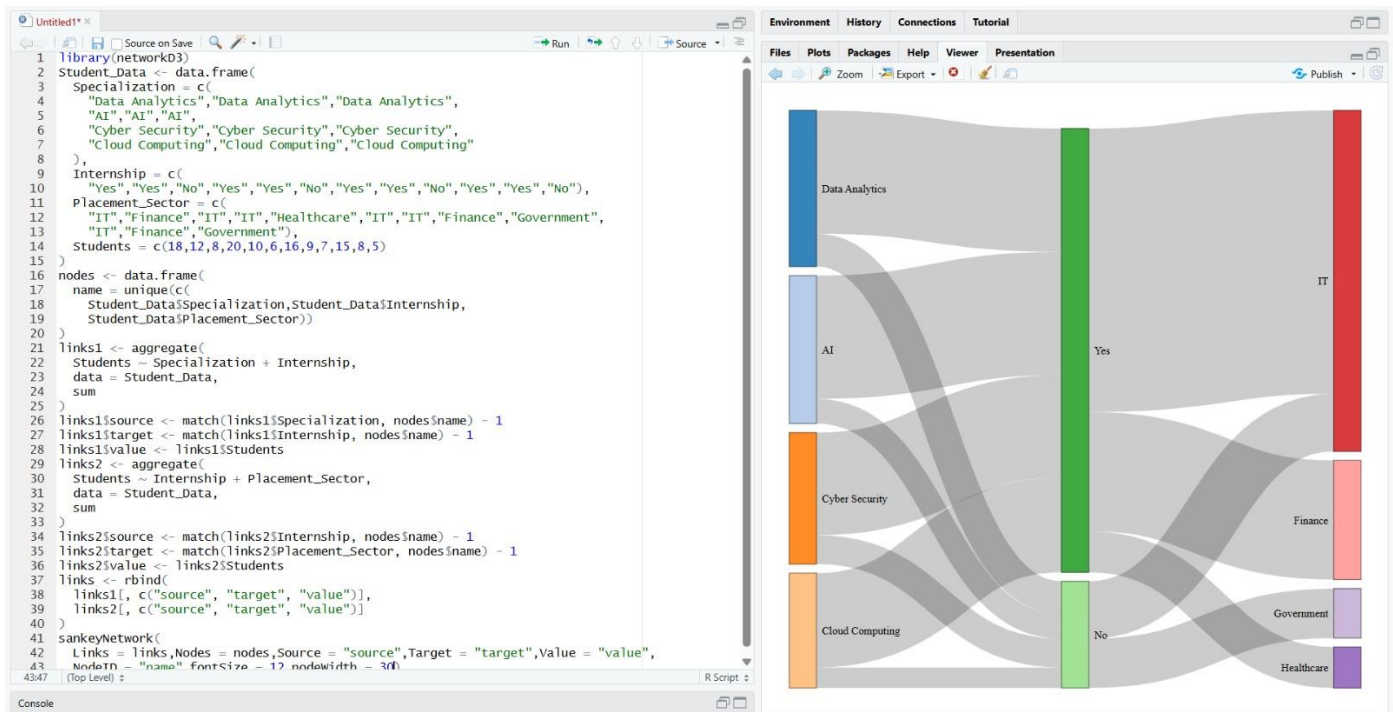
1. Create a Sankey diagram showing Specialization → Internship → Placement_Sector, with flow width based on Students.

Answer:

The Sankey diagram shows the flow:

Specialization → Internship → Placement Sector

The width of each flow represents the number of students



2. Create a parallel sets visualization for the same categorical relationships.

Answer:

The parallel sets visualization shows the relationship between specialization, internship status and placement sector.

3. Identify the major flows and compare the placement patterns of students with and without internships.

Answer:

- **AI → Yes → IT** is the largest individual flow with **20 students**.
- **Data Analytics → Yes → IT** has **18 students**.
- **Cyber Security → Yes → IT** has **16 students**.
- **Cloud Computing → Yes → IT** has **15 students**.
- Students with internships are mainly placed in **IT**.
- Students without internships are placed in **IT** and **Government**.

The student counts above come directly from the dataset in the assessment document.

4. Explain which visualization communicates the flow structure more clearly and why.

Answer:

The **Sankey diagram** communicates the flow structure more clearly because:

- It shows the direction of movement between categories.
- Flow width represents the number of students.
- Major and minor flows can be identified easily.
- It clearly shows the relationship between specialization, internship and placement.
- It is useful for understanding the overall career pathway.

The **parallel sets visualization** is useful for comparing several categorical dimensions at the

same time.

5. Discuss one way the chart could become misleading if proportions are represented using inappropriate scales, colors, or labels.

Answer:

The chart can become misleading if inappropriate **scales, colors or labels** are used.

For example:

- An incorrect flow width can make a small group appear larger.
- Misleading colors can make some categories appear more important.
- Incorrect labels can cause the viewer to misunderstand the flow.
- Unequal visual scaling can distort the actual proportions.
- Poorly chosen colors can make different categories difficult to distinguish.

Dataset:

Student_Group	Specialization	Internship	Placement_Sector	Students
G1	Data Analytics	Yes	IT	18
G2	Data Analytics	Yes	Finance	12
G3	Data Analytics	No	IT	8
G4	AI	Yes	IT	20
G5	AI	Yes	Healthcare	10
G6	AI	No	IT	6
G7	Cyber Security	Yes	IT	16
G8	Cyber Security	Yes	Finance	9
G9	Cyber Security	No	Government	7
G10	Cloud Computing	Yes	IT	15
G11	Cloud Computing	Yes	Finance	8
G12	Cloud Computing	No	Government	5

Expected Student Deliverables

- R script (.R) containing data creation/import, data preparation, analysis, and visualization code.
- All required plots with meaningful titles, axis labels, legends, and appropriate scales.
- A short interpretation for each visualization.
- A brief comparison of alternative visualization methods where requested.
- A concluding statement identifying important patterns, differences, or potential misleading visual encodings.

Suggested Assessment Criteria

Criterion	Weight
Dataset preparation	20%
Correct R programming and use of visualization packages	25%
Appropriate choice and construction of visualizations	25%
Interpretation and analytical reasoning	20%
Clarity, labeling, and presentation	10%